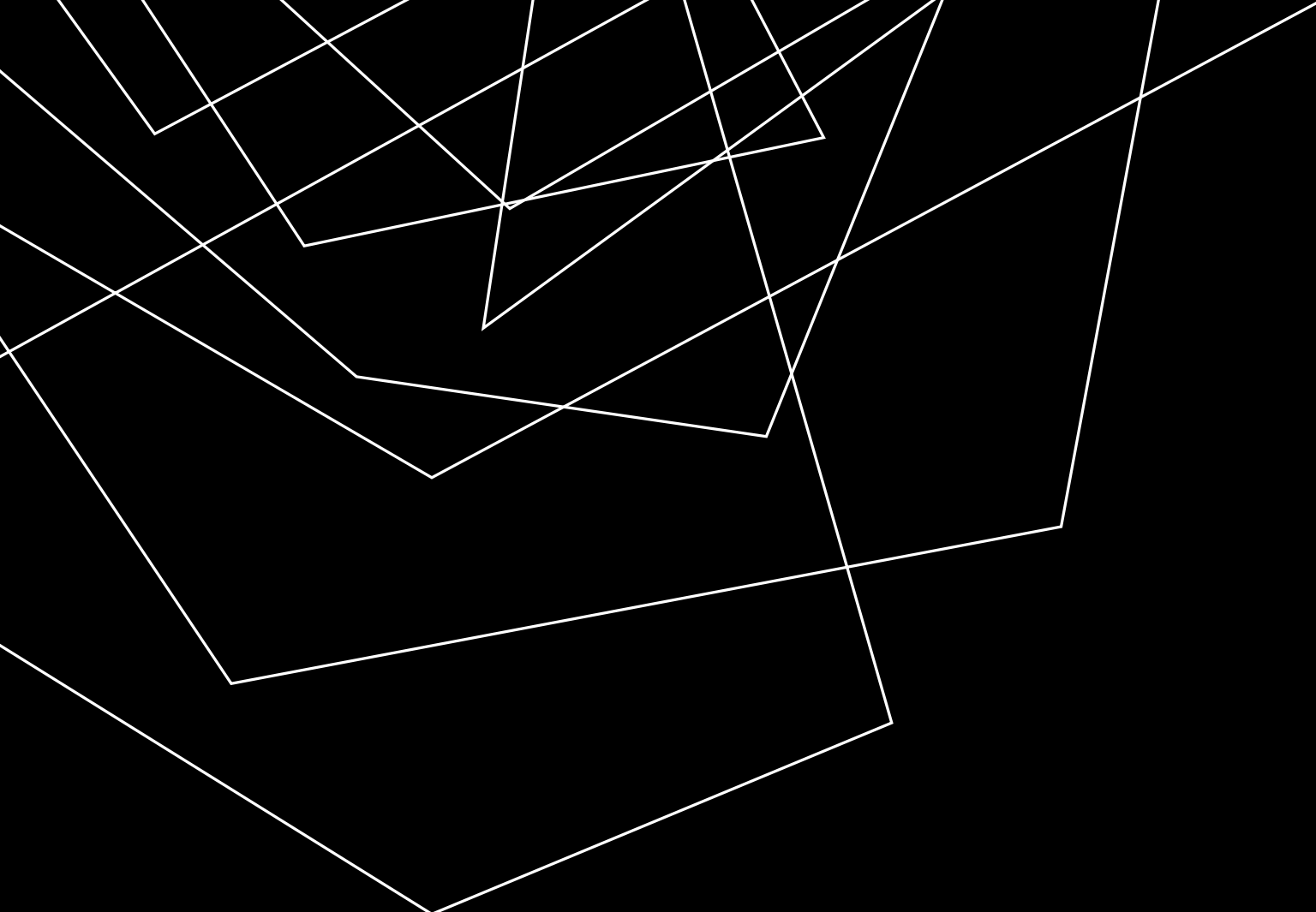




POP RAG SEARCH ENGINE

ZARINA BEISENBAYEVA

THE BEGINNING OF THE JOURNEY



- Wanted to build a RAG Application on Sabrina Carpenter's songs
- Turning point:
 - The comment from one expert "If you want ot build a good music retrieval, think of expanding your dataset"

THE PROBLEM

- After chunking and indexing Sabrina's songs(89), I had only ~300 chunks for the whole corpus
- Indeed, the experience would be quite boring...

THUS, I decided to expand my dataset focusing not only on Sabrina, but on the part of the music industry



WHAT DO I HAVE
NOW?



THE GOAL

- A Retrieval-Augmented Generation (RAG) system over song lyrics
- Goal: let users ask natural-language questions
- For example: the user enters *"songs about missing toxic ex"* and receive both relevant lyric snippets and a concise, cited answer explaining why those songs match

POP RAG Search Engine

Enter your query:

songs about missing toxic ex

Number of suggestions you would like to receive:



Search and Answer

Answer: ↗

A BIT ABOUT DATA

- 7 artists
- Each artist's CSV has columns:
 - Artist
 - Title
 - Album
 - Year
 - Lyric
- After chunking, I had 4235 chunks:
 - Dua Lipa has 495 indexed chunks
 - Ariana Grande has 859 chunks from
 - Billie Eilish has 174 chunks
 - Sabrina Carpenter has 396 chunks
 - Maroon 5 has 502 chunks
 - Ed Sheeran has indexed 775 chunks
 - Justin Bieber has indexed 1034 chunks

PREPROCESSING & CLEANING

- Filtering unwanted versions:
 - Removed remixes, live/acoustic versions via regex on Title
 - Keywords: Remix, Live, Acoustic, Demo, Cover, and many more...
- Output:
 - Cleaned CSVs stored in “Cleaned_csvs” folder

```
    ]  
    escaped = [re.escape(k) for k in keywords]  
    pattern = rf"({'|'}.join(escaped)}")  
  
    output_folder = os.path.join(csv_folder, "Cleaned_csvs")  
    os.makedirs(output_folder, exist_ok=True)
```

MY INDEXING PART

1. Chunking Process:

- Read each raw lyric file (one per song)
- Split into overlapping windows of ~200 tokens (20–30 token overlap)
- Pair each chunk with an integer index (1, 2, 3, ...), read the artist name from, and combine into:
 - Sabrina_Carpenter_1
- Store metadata: artist, song title, album, year, chun_index in a song

2. Embedding & Ingestion into ChromaDB:

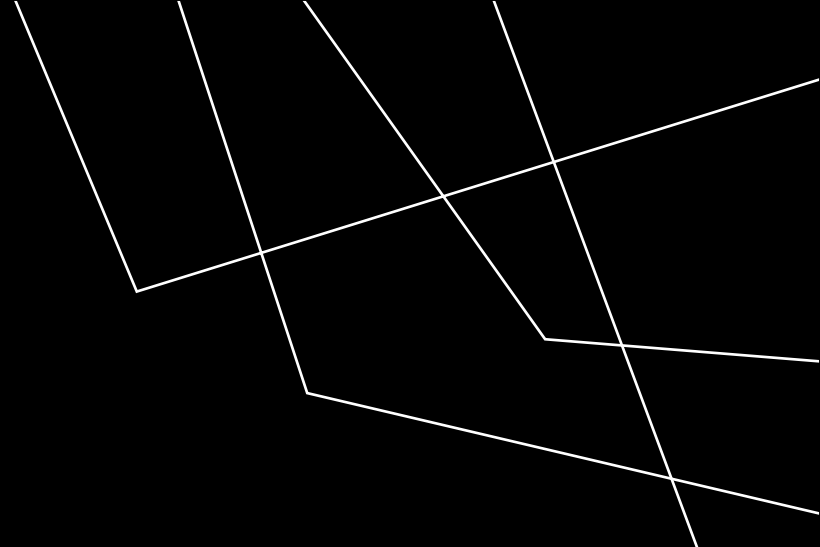
- Used ChromaDB Client for storing vectors
- HNSW graph with cosine similarity

```
docs      = [chunk.page_content for chunk in chunks]
ids       = [f"{chunk.metadata['artist'].replace(' ', '_')}_{i}"
             for i, chunk in enumerate(chunks, start=1)]
metadatas = [chunk.metadata for chunk in chunks]
collection.add(
    documents=docs,
    ids=ids,
    metadatas=metadatas
)
total_indexed += len(chunks)
print(f"Indexed {len(chunks)} chunks from {os.path.basename(csv_path)}")
```

DOCUMENT FORMAT

- Each “document” is a chunk of lyrics (≈200 tokens, overlapping windows)
- Metadata per chunk includes:
 - artist (“Billie Eilish”)
 - song_title (“when the party’s over”)
 - song_album (“WHEN WE ALL FALL ASLEEP, WHERE DO WE GO?”)
 - song_year (2019)
 - chunk_index(11)

```
for idx, doc in enumerate(docs_for_this_song, start=1):  
    doc.metadata.update({  
        "artist":      artist,  
        "song_title":  song_title,  
        "song_album":  song_album,  
        "song_year":   song_year,  
        "chunk_index": idx  
    })  
    chunks.append(doc)
```



```
{'ids': ['Billie_Eilish_10'], 'embeddings': None, 'documents': ["don't be that way fall apart twice a day i just wish you could feel what you say show never tell but i know you too well kind of mood that you wish you could sell if teardrops could be bottled there'd be swimming pools filled by models told a tight dress is what makes you a whore if i love you was a promise would you break it if you're honest tell the mirror what you know she's heard before i don't wanna be you anymore"], 'uris': None, 'included': ['metadatas', 'documents'], 'data': None, 'metadatas': [{'artist': 'Billie Eilish', 'song_album': 'dont smile at me', 'song_year': 2017, 'chunk_index': 1, 'song_title': '\u200bidontwannabeyouanymore'}]}\n{'ids': ['Billie_Eilish_11'], 'embeddings': None, 'documents': ["hands hands getting cold losing feeling is getting old was i made from a broken mold hurt i can't shake we've made every mistake only you know the way that i break if teardrops could be bottled there'd be swimming pools filled by models told a tight dress is what makes you a whore if i love you was a promise would you break it if you're honest tell the mirror what you know she's heard before i don't wanna be you i don't wanna be you i don't wanna be you anymore"], 'uris': None, 'included': ['metadatas', 'documents'], 'data': None, 'metadatas': [{'artist': 'Billie Eilish', 'song_title': '\u200bidontwannabeyouanymore', 'song_year': 2017, 'chunk_index': 2, 'song_album': 'dont smile at me'}]}
```

HOW DOES RETRIEVAL WORK?

User Query → Embedding → Nearest Neighbors

- Ask ChromaDB for top K nearest lyric chunks(K = 3)
- Include documents and metadatas in the query response

Build RAG Prompt + Call LLM

- XML-style wrapper with search results
- GPT-4 for answer generation and song recommendations

Contextual Expansion (“prev + current + next”)

- During retrieval, retrieval model will focus entirely on finding the most relevant search results by analysing neighbours
- The generation model will focus on generating "good" responses based on the retrieved data

Build Demo Version using
Streamlit!!!



CHALLENGES FACED

- The initial Sabrina feature has been postponed
- Hard-coding filtering to remove alternate versions (remix, live, acapella, etc.)
- Intended to show full chunk details in “detailed” answers, but the decoupling logic isn’t working as expected
- Requesting the top 3 songs sometimes yields 4 or 5 results...

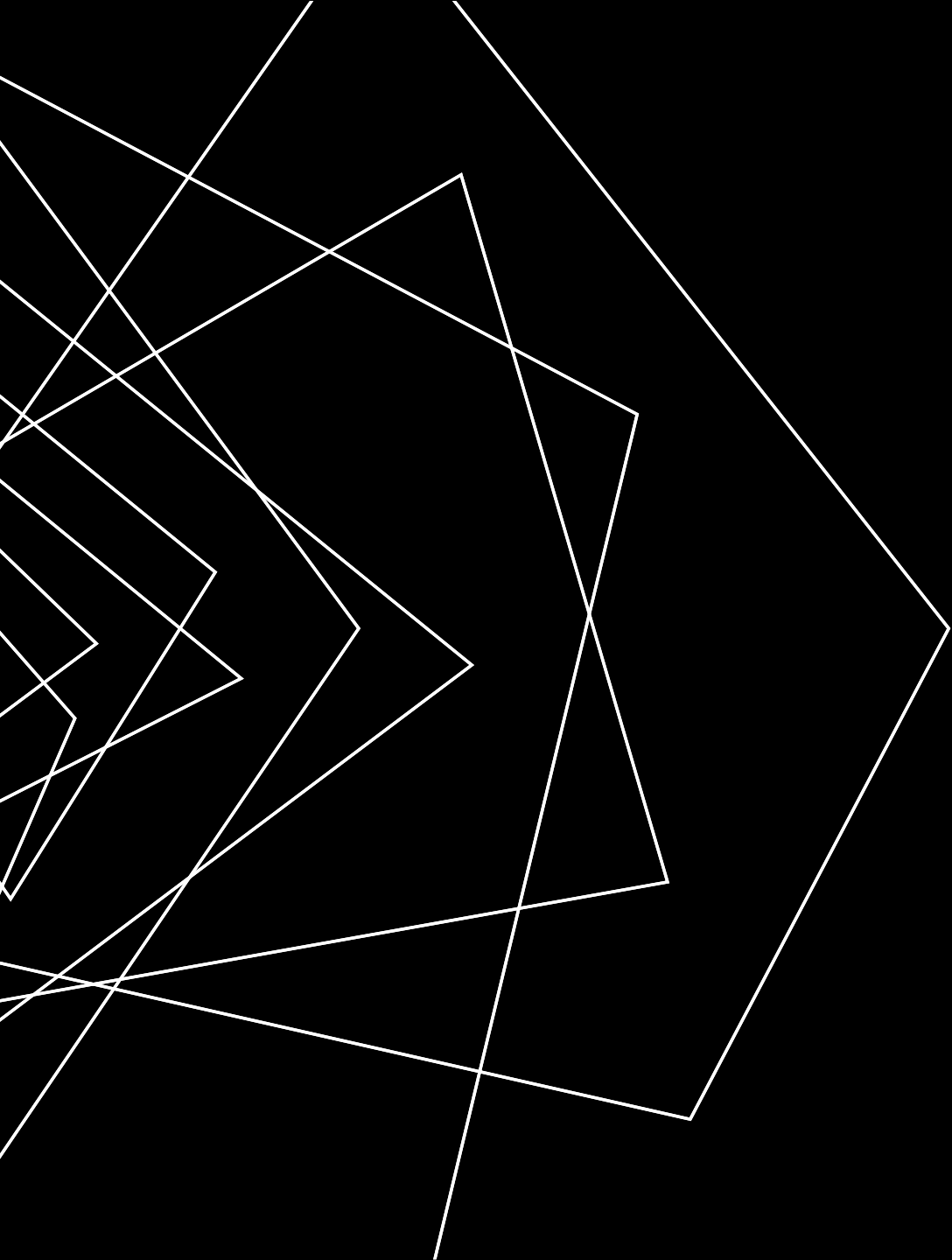
1. "Please Please Please" by Sabrina Carpenter from the album "Short n' Sweet (Deluxe), 2024 - Sabrina thrillingly voices a plea to not cause her any more pain and asserts her wish not to be embarrassed again. Her good humour masks the deeper reality in the lyrics, "Don't bring me to tears when I just did my makeup so nice. Heartbreak is one thing, my ego's another" - [Source: "Please Please Please", Sabrina Carpenter, Short n' Sweet (Deluxe), 2024]
2. "Don't Start Now" by Dua Lipa from the album "Future Nostalgia, 2019 - Dua Lipa navigates through the journey of moving on and warns the ex not to reappear in her life through her lyrics, "if you don't wanna see me dancing with somebody, if you wanna believe that anything could stop me, don't show up. Don't come out" - [Source: "Don't Start Now", Dua Lipa, Future Nostalgia, 2019]
3. "New Rules" by Dua Lipa from the album "Dua Lipa, 2017 - This song sees Dua Lipa setting new rules for herself following a break up. The catchy chorus lays down three rules for how to move on from an ex who, she feels, is not treating her right - [Source: "New Rules", Dua Lipa, Dua Lipa, 2017]
4. "Nonsense" by Sabrina Carpenter from the album "emails i can't send fwd:, 2022 - In this song Carpenter talks about forgetting her ex and how she loses her calmness around her new interest, narrated in the words, "lookin' at you got me thinkin' nonsense. Cartwheels in my stomach when you walk in" - [Source: "Nonsense", Sabrina Carpenter, emails i can't send fwd:, 2022]
5. "I'm Glad I'm Not You" by Ed Sheeran from "Want Some? - EP" - This song expresses Ed Sheeran's relief of not being in the place of his ex, while still acknowledging the feelings of confusion and the lingering effects of the relationship - [Source: "I'm Glad I'm Not You", Ed Sheeran, Want Some? - EP]

FUTURE IMPROVEMENTS

- Update the golden list to evaluate IR model
- Experiment with LLM temperature
- Add semantic filters (genre, mood)
- Alternative embedding models (audio embeddings)



GITHUB LINK:
[HTTPS://GITHUB.COM/ZARINESS00/IR-IE-COURSE](https://github.com/ZARINESS00/IR-IE-COURSE)



THANK YOU

Zarina Beisenbayeva